Project Report - COGS 189

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Abstract

This project studies how EEG signals from motor imagery can be used to control a simple brain-computer interface (BCI). The goal is not only to classify EEG signals, but also to test whether these predictions can support a usable typing system. Two approaches are compared: a classical method using CSP + LDA and a deep learning model (EEGNet). The results show that CSP + LDA achieves higher accuracy on this dataset, while EEGNet performs worse due to limited data. A simulated typing interface is then used to evaluate usability. The results show that raw predictions are unstable, but applying confidence thresholds and smoothing improves control. This project highlights that both model accuracy and output stability are important for real BCI systems.

1 Introduction

EEG-based brain-computer interfaces aim to translate brain activity into control signals. One common approach is motor imagery, where users imagine moving different body parts to generate distinct EEG patterns. These signals can then be used to control external systems.

Many studies focus on improving classification accuracy. However, in real applications, accuracy alone is not enough. Even a model with good accuracy may produce unstable outputs, which makes it difficult for users to control a system.

Previous work has shown that CSP combined with LDA is effective for motor imagery classification, especially on small datasets. EEGNet is a compact deep learning model designed for EEG decoding, but typically requires more data to perform well. In this project, I compare these two approaches and extend the problem by evaluating how well their predictions can support a usable control system.

2 Method

I use the BCI Competition IV Dataset 2a, which contains EEG recordings from motor imagery tasks. The data includes 22 channels sampled at 250 Hz. In this project, I focus only on left and right hand imagery to form a binary classification problem.

The EEG signals are first filtered between 8 and 30 Hz to capture motor-related activity. I then extract events and segment the data into trials from 0 to 4 seconds. These trials are used as input to the models.

Two models are implemented. The first is a classical approach using Common Spatial Patterns (CSP) for feature extraction and Linear Discriminant Analysis (LDA) for classification. The second is EEGNet, a convolutional neural network designed to learn both spatial and temporal features directly from EEG data.

To evaluate the system beyond classification, I also design a simulated typing interface. The model outputs are mapped to cursor movement and selection actions, allowing us to test how well the predictions translate into control.

3 Experiments

I first analyze the EEG signals to verify that they contain useful information. The power spectral density (PSD) shows strong activity in the 8–30 Hz range, which is consistent with motor imagery signals.

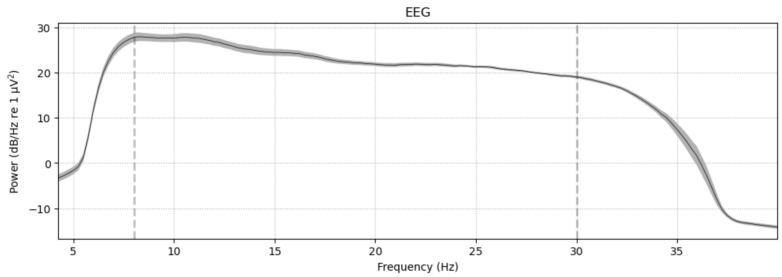


Figure 1: Power spectral density of EEG signals showing strong activity in the 8–30 Hz range.

I also examine signals from motor-related channels (C3, Cz, C4). These channels show different patterns for left and right imagery, indicating that the data contains discriminative spatial information.

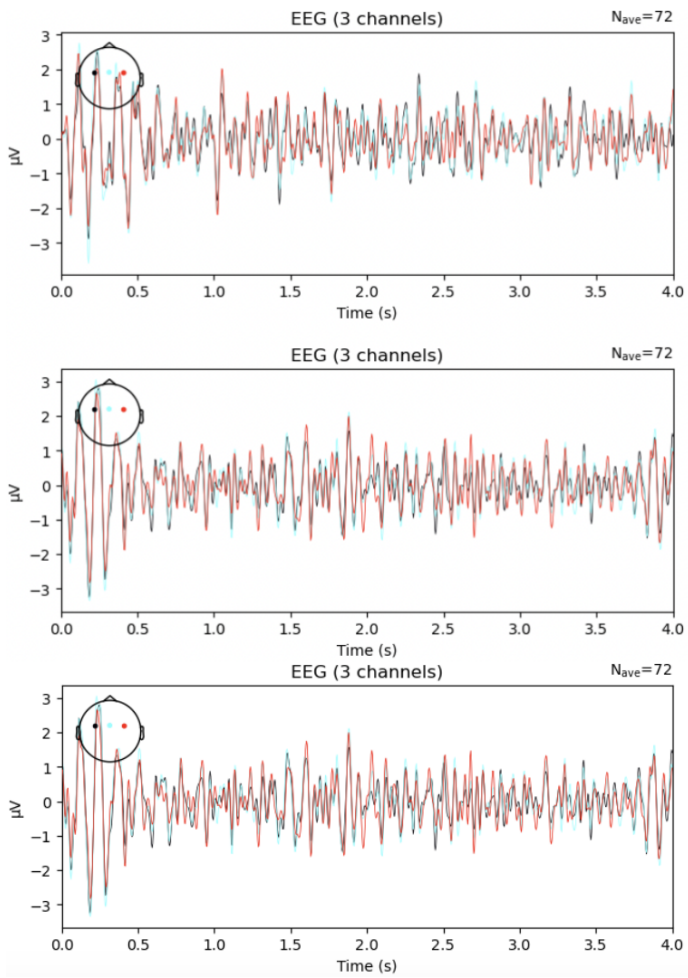


Figure 2: Average EEG signals at channels C3, Cz, and C4 for left and right motor imagery.

Figure 2 shows the average EEG signals from channels C3, Cz, and C4 during left and right motor imagery. Clear differences can be observed between the two classes, especially in the motor cortex

regions. These differences indicate that the EEG signals contain discriminative patterns that can be used for classification. This supports the feasibility of using machine learning models to decode user intent from brain signals in this project.

I then train the CSP + LDA model using cross-validation. The model achieves an average accuracy of around 81 percent, indicating strong performance on this dataset.

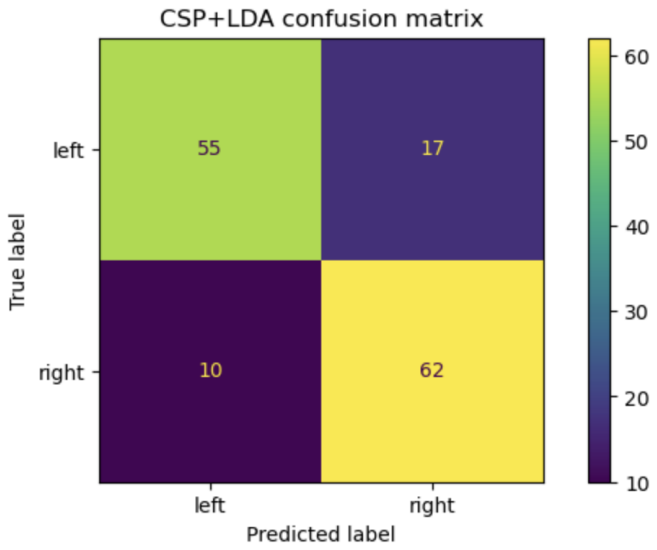


Figure 3: Confusion matrix for CSP + LDA classification.

Figure 3 presents the confusion matrix of the CSP + LDA model. The model correctly classifies 117 out of 144 trials, achieving an accuracy of approximately 81 percent. Most predictions fall along the diagonal, indicating reliable separation between left and right motor imagery. This result demonstrates that the model provides a strong and stable foundation for controlling the typing interface.

Next, I train EEGNet. The model converges during training, but achieves lower accuracy of around 61 percent. The loss curve shows that the model is learning, but may not have enough data to reach higher performance.

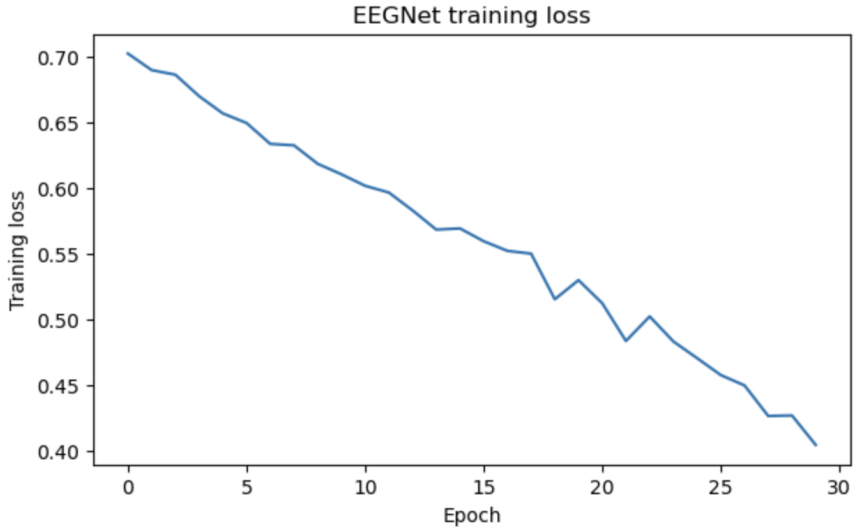


Figure 4: Training loss curve for EEGNet.

Figure 4 shows the training loss of the EEGNet model over epochs. The loss steadily decreases from around 0.70 to 0.41, indicating that the model is learning meaningful patterns from the data. However, compared to CSP + LDA, EEGNet does not achieve higher performance in this task. This suggests that simpler models may be more effective for this dataset and application.

To understand what the model is learning, I compute saliency maps. The saliency map shows structured activity over central and nearby channels, including regions near the motor electrodes.

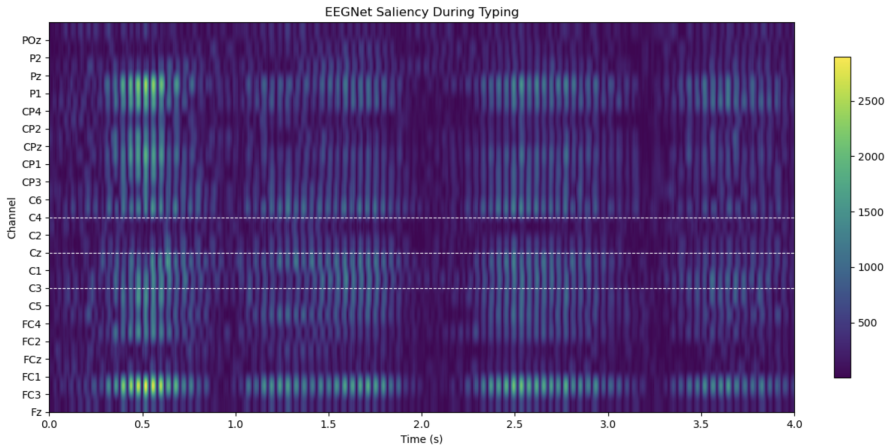


Figure 5: Saliency map highlighting important EEG channels.

Figure 5 illustrates the saliency map of the EEGNet model. The model focuses on regions around the central motor areas, including channels near C3 and C4, which are known to be related to motor imagery. This provides insight into how the model makes decisions and confirms that it is using meaningful brain signal features rather than noise.

4 Results and Analysis

Although CSP + LDA achieves higher classification accuracy, the typing simulation reveals a different challenge. When raw predictions are used directly, the system produces unstable behavior with many incorrect actions.

I test three control strategies: raw predictions, confidence thresholding, and temporal smoothing. The results show that thresholding reduces noise, and smoothing further improves stability.

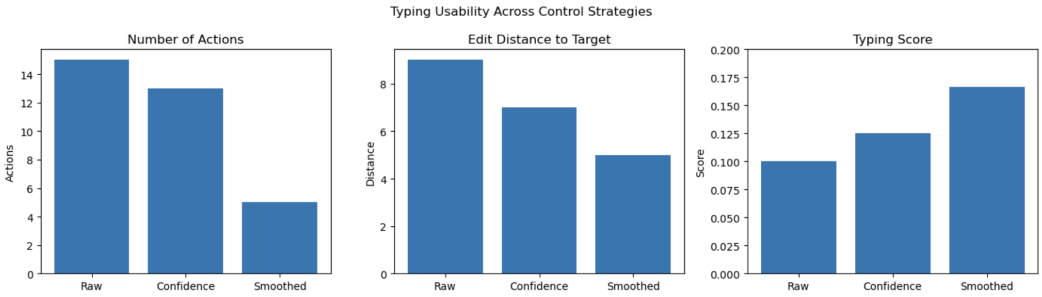
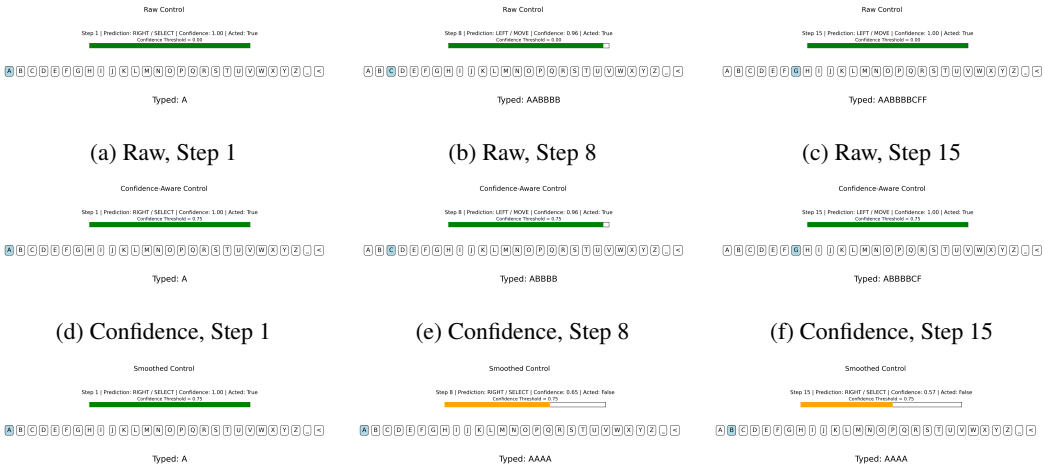


Figure 6: Comparison of typing usability metrics under different control strategies.

Figure 6 compares typing usability metrics across different control strategies. Raw control produces the highest number of actions and the largest edit distance, indicating unstable and inefficient behavior. Confidence-aware control reduces unnecessary actions by filtering out low-confidence predictions, leading to moderate improvement. Smoothed control achieves the best performance, with the fewest

actions and the lowest edit distance. These results show that post-processing significantly improves usability, even when using the same classifier.

To further visualize system behavior, I created an animated typing interface. This shows how predictions evolve over time and how the cursor moves.



(a) Raw, Step 1 (b) Raw, Step 8 (c) Raw, Step 15

(d) Confidence, Step 1 (e) Confidence, Step 8 (f) Confidence, Step 15

(g) Smoothed, Step 1 (h) Smoothed, Step 8 (i) Smoothed, Step 15

Figure 7: Comparison of typing behavior under different control strategies.

Figure 7 shows example frames from the typing simulation under different control strategies. Raw control produces rapid and unstable character updates, often leading to incorrect outputs. Confidence-aware control reduces some of this noise by limiting actions to high-confidence predictions. Smoothed control produces the most stable typing behavior, with fewer unnecessary changes and more consistent progress toward the target word. This figure highlights the practical impact of control strategies on user experience.

This shows that building a usable BCI requires not only accurate models, but also stable and reliable control over time. There are several possible improvements to this system. First, adaptive confidence thresholds could be used to adjust the decision boundary based on signal quality, which may improve reliability. Second, temporal models such as recurrent neural networks could better capture dependencies across time and produce more stable predictions.

Conclusion

This project shows that we can use EEG signals from motor imagery to control a simple BCI system, but it also shows that accuracy alone is not enough.

The CSP + LDA model performed better than EEGNet, reaching about 81 percent accuracy. This likely happens because our dataset is not very large. CSP is designed for small EEG datasets and works well at extracting useful spatial patterns. EEGNet, on the other hand, usually needs more data to learn effectively, so its performance is lower here. More importantly, when I moved from classification to actual control (typing simulation), I saw a new problem. Even if the model is accurate, the output can still be unstable. The raw predictions caused too many unwanted actions. When I added confidence thresholds and smoothing, the system became much easier to control. This shows that building a BCI is not just about getting high accuracy. I also need to think about how the predictions behave over time and how the user interacts with the system.

There are still some limitations. I only used a small dataset and a simple left vs right task. Also, everything was done offline, so real-time performance may be different. In the future, I can improve this system by testing it in real time, adapting it to each user, and improving the deep learning model with more data. Overall, this project shows that a good BCI system needs both good models and stable control behavior.